

5a	weight provides the centripetal force (or acceleration of free fall is centripetal acceleration) $9.81 = 0.130 \times \omega^2$ $\omega = 8.687 = 8.7 \text{ rad s}^{-1}$	B1 M1 A0
5b	force in cord – weight = centripetal force $T - W = mr\omega^2$ force constant $k = 5.0/0.018$ $(L - 0.013) \times 5.0/0.018 - 5.0 = 5.0/9.81 \times L \times 8.7^2$ $L = 0.172 \text{ m} = 17.2 \text{ cm}$	C1 C1 A1

6ai	When light intensity is maximum, resistance of LDR = 1200Ω	C1
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